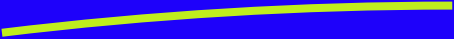


LEGO: VALUABLE FOR COLLECTORS?



BUS32120: Data Analysis with Python & SQL Final



INTRODUCTION

INTRODUCTION



A.

Target audience:

LEGO collectors and secondary market investors looking to purchase LEGO sets as an investment.

B.

Key Question:

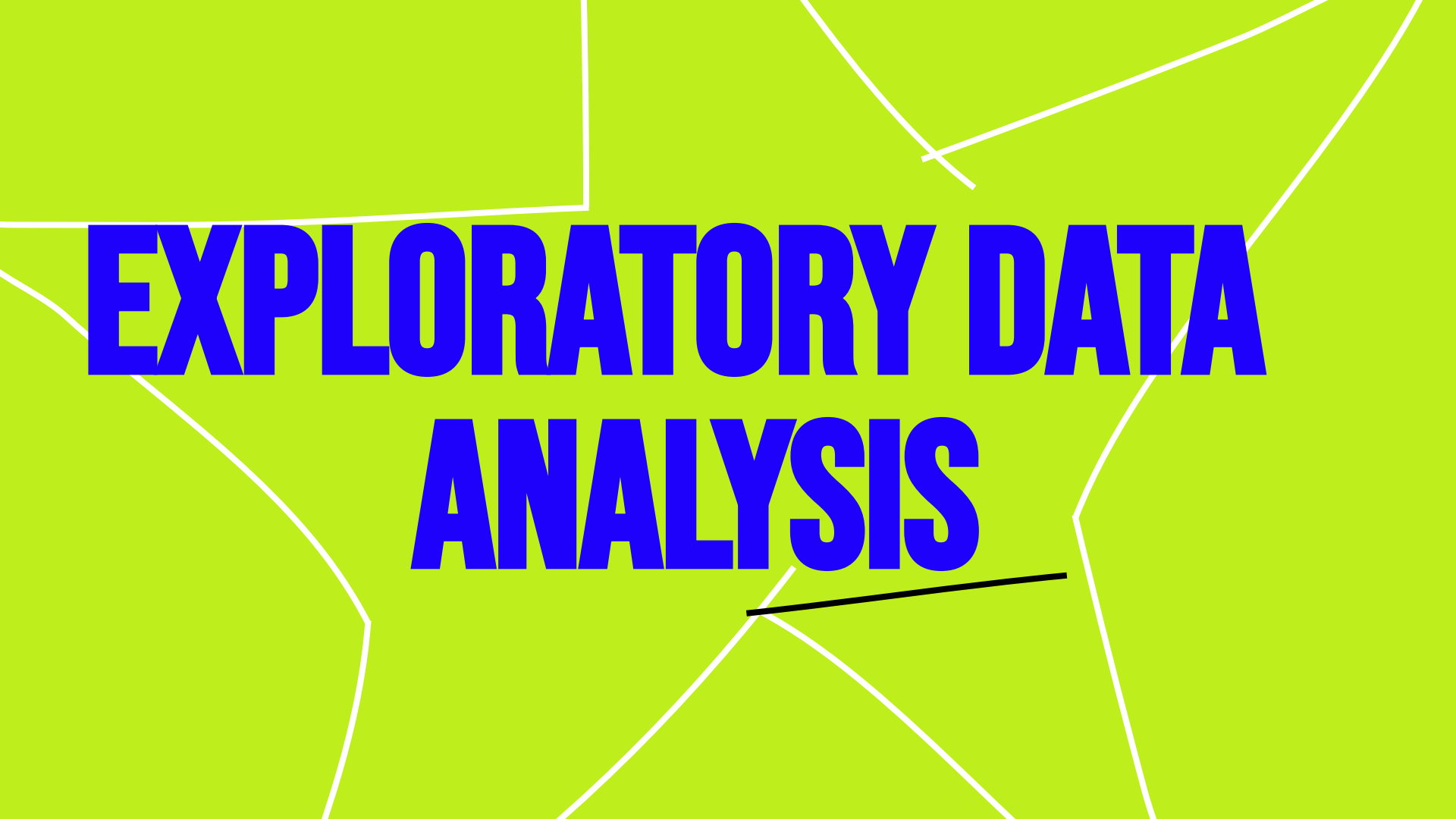
What physical set characteristics make a LEGO set a good investment and can we predict price appreciation before a set is retired?

C.

Data Sources:

- Rebrickable (physical set characteristics, set numbers, themes, etc).
- BrickLink (secondary market pricing at time of retirement, and one year post retirement).

EXPLORATORY DATA ANALYSIS

The background is a solid yellow-green color. It features several thin, white, irregular lines that intersect and curve across the frame, creating a sense of movement and abstraction. A single, thicker black line runs diagonally from the bottom left towards the center right, positioned below the main text.

DISTRIBUTION OF PRICE APPRECIATION

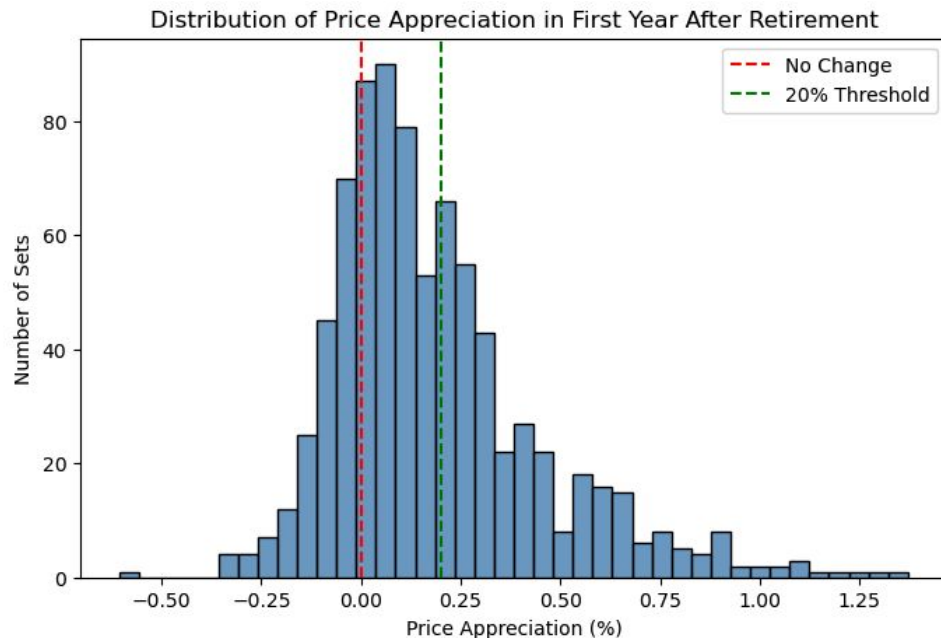
MOST LEGO SETS DON'T APPRECIATE, BUT A FEW DO SIGNIFICANTLY

A.

Most sets cluster around 0% appreciation. Some sets actually lose value after retirement.

B.

Only a subset of sets cross the 20% threshold to be considered a "good investment", making set selection critical.



PERCENTAGE OF GOOD INVESTMENTS BY THEME

NICHE THEMES OUTPERFORM ICONIC FRANCHISES.

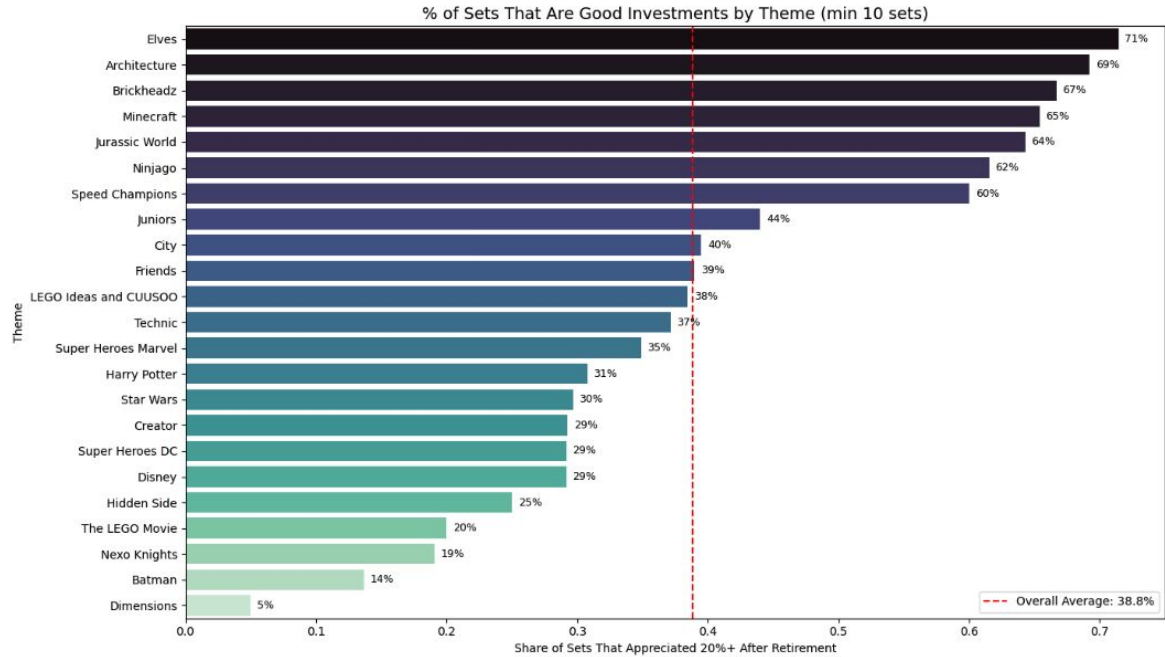
A.



Elves, Architecture, and Brickheadz top the chart, all niche, collectible-oriented themes with limited set releases.

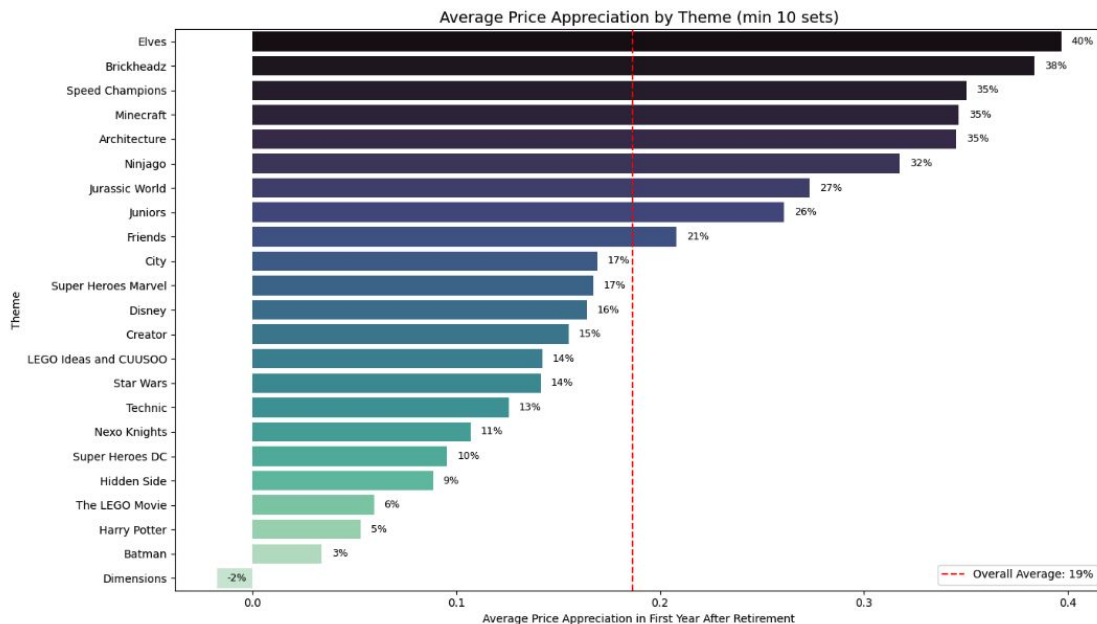
B.

Star Wars (30%) and Harry Potter (32%) fall well below the overall average of 38.8%, despite being LEGO's most recognizable franchises.



AVERAGE PRICE APPRECIATION BY THEME

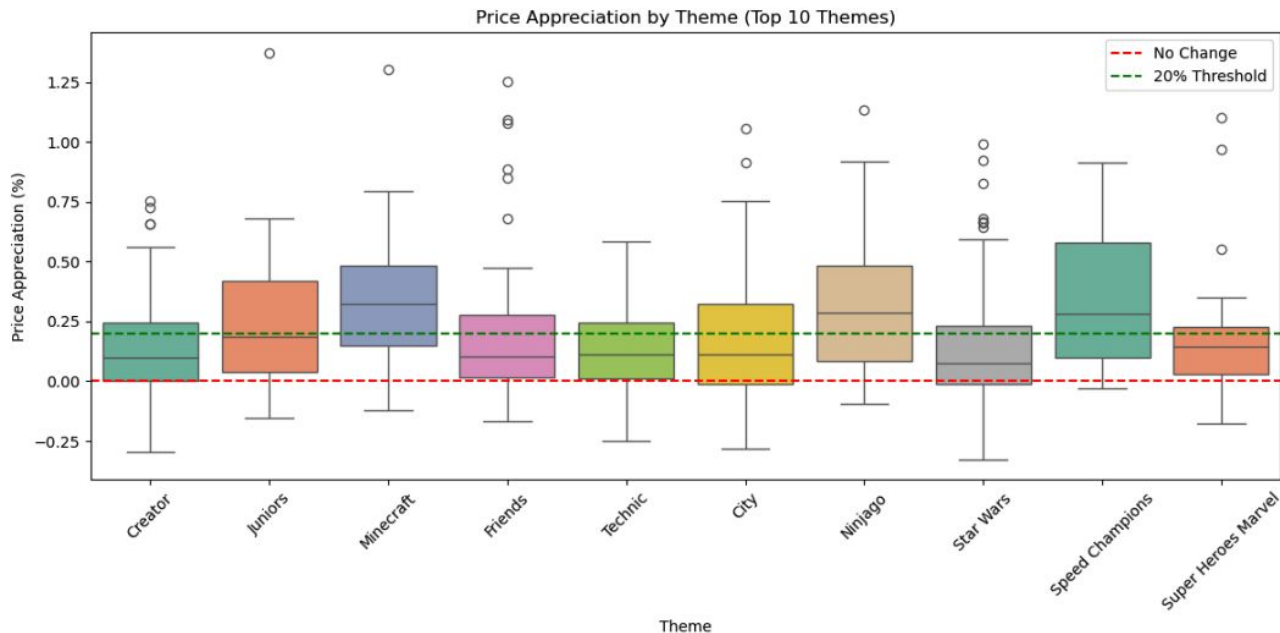
	parent_theme_name	priced_sets	avg_price_appreciation
0	Elves	14	0.396923
1	Brickheadz	15	0.383571
2	Speed Champions	25	0.350261
3	Minecraft	26	0.346416
4	Architecture	13	0.345221
5	Ninjago	65	0.317671
6	Jurassic World	14	0.273406
7	Juniors	25	0.260733
8	Friends	59	0.207917
9	City	81	0.169215
10	Super Heroes Marvel	43	0.167250
11	Disney	24	0.164104
12	Creator	41	0.154990
13	LEGO Ideas and CUUSOO	13	0.142081
14	Star Wars	118	0.141336
15	Technic	35	0.125710
16	Nexo Knights	21	0.107035
17	Super Heroes DC	24	0.095486
18	Hidden Side	16	0.088550
19	The LEGO Movie	15	0.059512
20	Harry Potter	13	0.053139
21	Batman	22	0.033989
22	Dimensions	20	-0.017519



**AVERAGE APPRECIATION
CONFIRMS THE THEME STORY,
WITH ONE KEY OUTLIER.**

PRICE APPRECIATION BY THEME

**PICKING THE
RIGHT THEME
ISN'T ENOUGH –
INDIVIDUAL SET
SELECTION
MATTERS.**

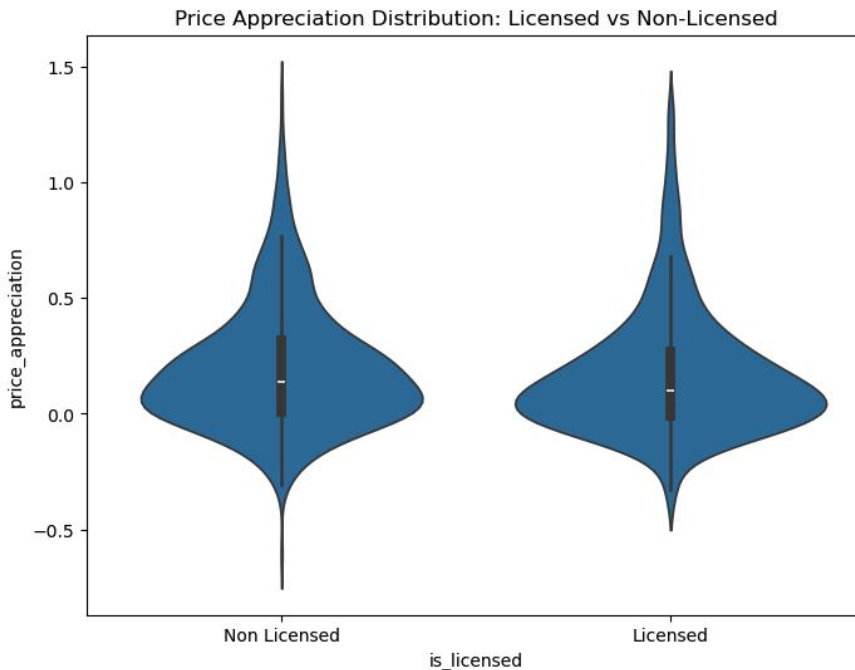


Every theme shows wide spread, meaning within any theme, some sets significantly outperform and others lose value.

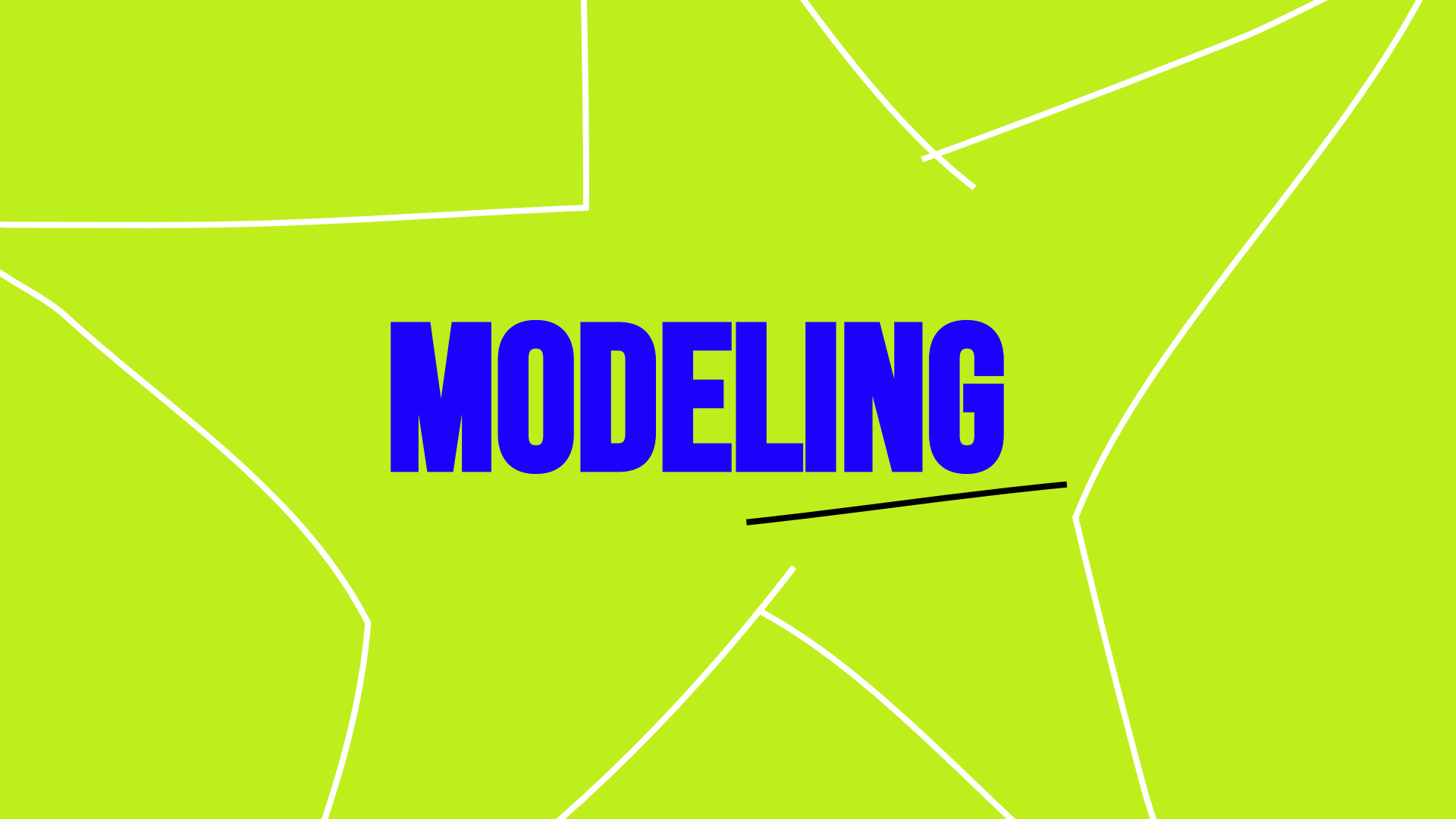
Even the best themes have losers. You can't just buy any set from a good theme and expect returns.

PRICE APPRECIATION BY THEME

**LICENSED VS.
NON-LICENSED
DOESN'T EXPLAIN
APPRECIATION EITHER.**



Both distributions overlap almost entirely, licensing alone is not a predictor of investment performance.

The background is a solid blue color with several white lines of varying thickness and curvature. These lines create a sense of movement and structure, some intersecting and others running parallel. The word 'MODELING' is centered in the upper half of the image.

MODELING

CAN A LEGO SET'S PRICE APPRECIATION BE PREDICTED BY THE SET'S KEY CHARACTERISTICS?

A. TARGET VALUE

Price Appreciation 1 year post set retirement

B. INPUT VARIABLES

- Number of parts
- Number of minifigs
- Ratio of parts to minifigs (set complexity)
- Color diversity
- Licensing

C. RESULTS

Low R2 of -0.007 and RMSE of 26% indicate model is underperforming. Model does not have strong predictive power, and price appreciation is difficult to predict with the input variables used.

Additional data on production / scarcity, collector signals (e.g. BrickLink watchlist count), awards, and social media / popularity could improve predictive power.

LINEAR REGRESSION RESULTS

R-squared: -0.007
RMSE: 0.2580

Feature Coefficients:

num_parts	+0.0001
minifig_count	-0.0127
color_diversity	+0.0041
is_licensed	-0.0379
parts_per_minifig	-0.0001

CAN A LEGO SET'S STATUS AS A GOOD / BAD INVESTMENT BE PREDICTED BY THE SET'S KEY CHARACTERISTICS?

A. TARGET VALUE

Binary variable denoting whether set is a good investment or not

B. INPUT VARIABLES

- Number of parts
- Number of minifigs
- Ratio of parts to minifigs (set complexity)
- Color diversity
- Licensing

C. RESULTS

Model's 68% accuracy is low - 61% of lego sets are poor investments, so predicting all sets as poor investments would have achieved similar accuracy

Precision of 1.0 is misleading as model only correctly predicted 3.7% of good investment sets and missed 96.3% of good investments (high false negative rate)

ROC-AUC of 0.519 shows model is barely better than random chance

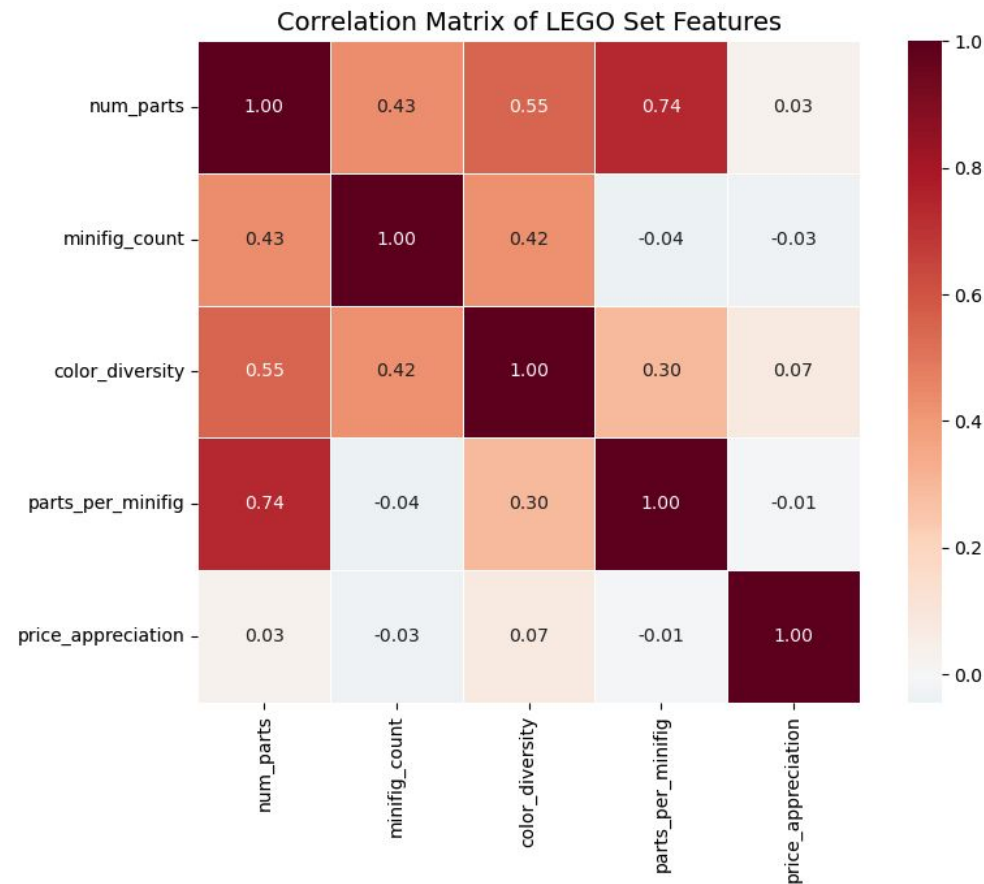
	Predicted Poor (0)	Predicted Good (1)
Actual Poor Investment (0)	109	0
Actual Good Investment (1)	52	2

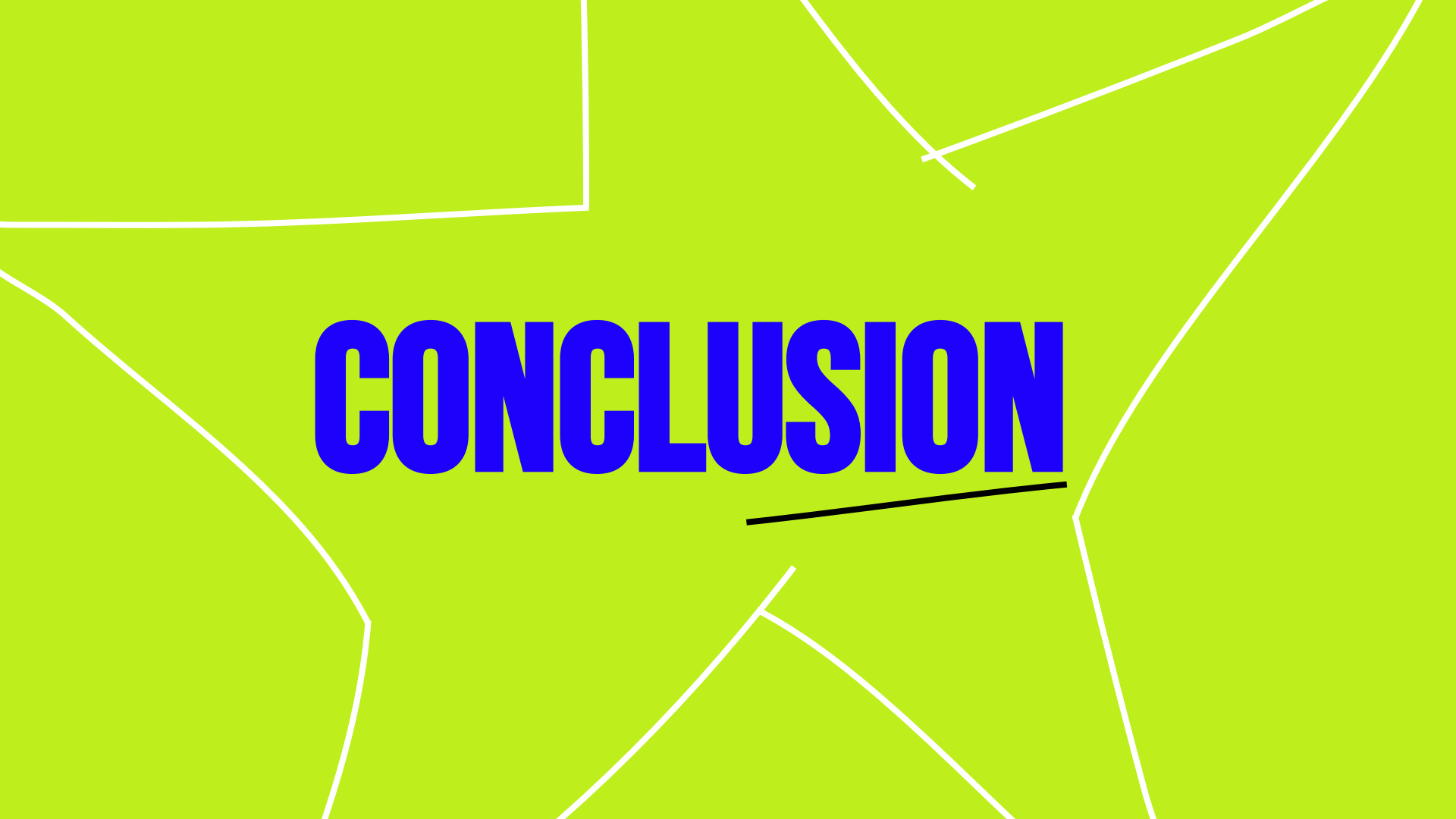
LOGISTIC REGRESSION RESULTS

Accuracy: 0.681
Precision: 1.000
Recall: 0.037
ROC-AUC: 0.519

CORRELATION MATRIX OF LEGO SET
FEATURES

DESIGN VARIABLES
RELATE TO EACH OTHER,
BUT DO NOT STRONGLY
PREDICT INVESTMENT
RETURNS.



The background is a solid blue color with several white, irregular, hand-drawn style lines scattered across it. These lines vary in length and orientation, some being straight and others curved, creating a dynamic and abstract pattern.

CONCLUSION

CONCLUSION



A.

Physical set characteristics do not predict price appreciation

B.

Investment performance varies across themes, with niche collectible themes outperforming franchises like Star Wars and Harry Potter

C.

Both models have low predictive power, reconfirming that set features are insufficient to predict returns

FOR LEGO INVESTORS, THE TAKEAWAY IS TO FOCUS ON THEME, EXCLUSIVITY / SCARCITY, AND COLLECTOR APPEAL INSTEAD OF PHYSICAL SET CHARACTERISTICS WHEN TRYING TO SELECT INVESTMENT LEGO SETS

APPENDIX

DISTRIBUTION OF SET SIZE (NUMBER OF PARTS)

LEGO'S PRODUCT STRATEGY APPEARS TO FOCUS ON ACCESSIBLE BUILDS RATHER THAN LARGE SETS.

A.

The number of pieces varies enormously:

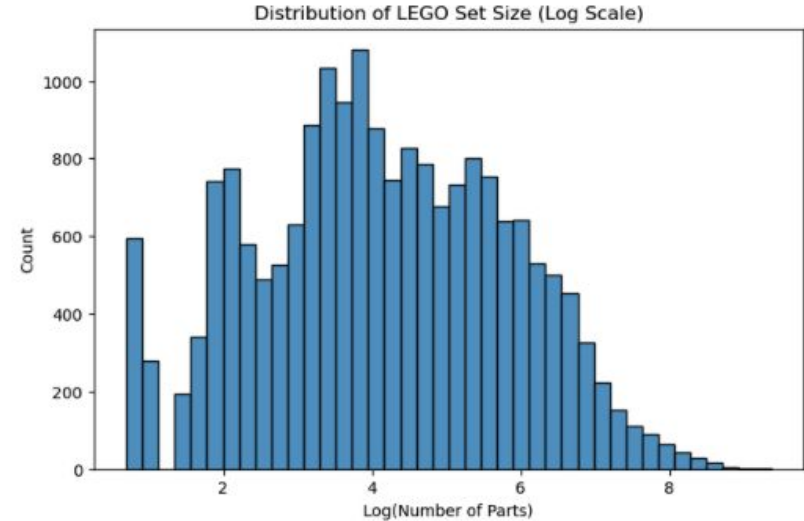
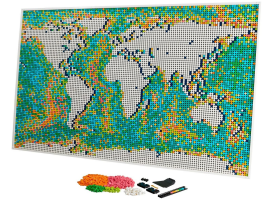
- Small sets: < 100 parts
- Large sets: > 5000 parts

B.

Most Lego sets are relatively small or medium sized. Very large sets exist but represent a small portion of the catalog.

C.

The max # parts a Lego can have is 11,695.



MOST COMMON LEGO THEMES

LEGO'S BALANCES CORE EVERGREEN THEMES AND LICENSED FRANCHISES.

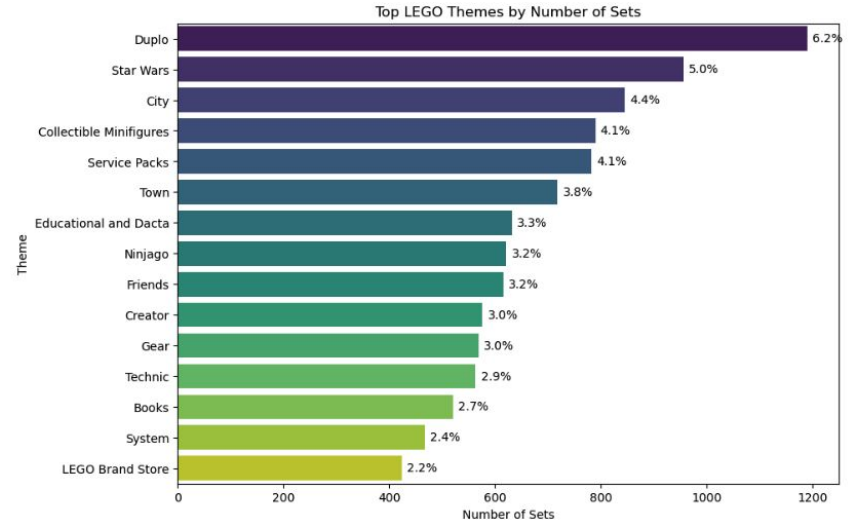
A.

Two of the largest number of sets are: City and Duplo



B.

Star Wars and Friends are the licensed franchises with the largest number of sets.



MINIFIGURES PER SET COUNTS

**MOST SETS ARE NOT
HEAVILY
CHARACTER-DRIVEN**

A.

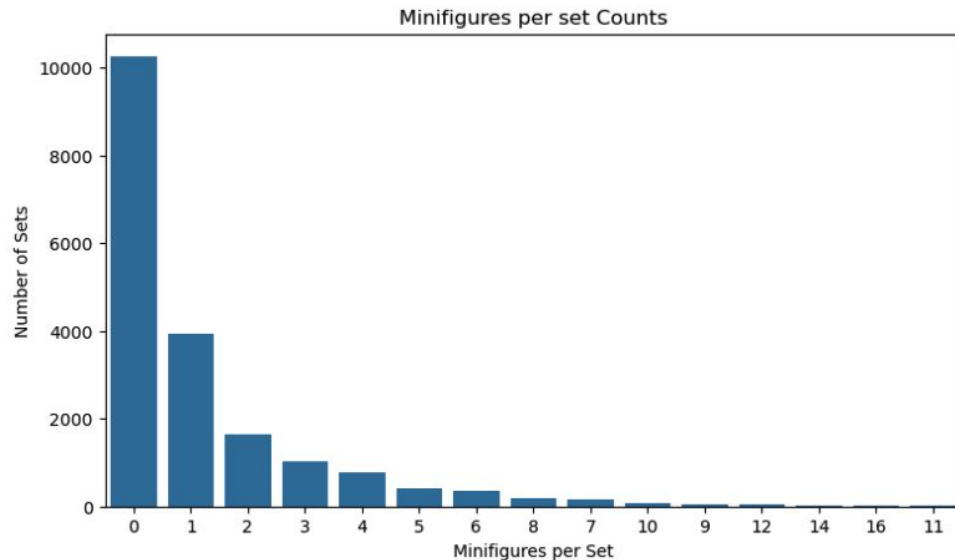
Most LEGO sets include:

- 0 to 4 minifigures

B.

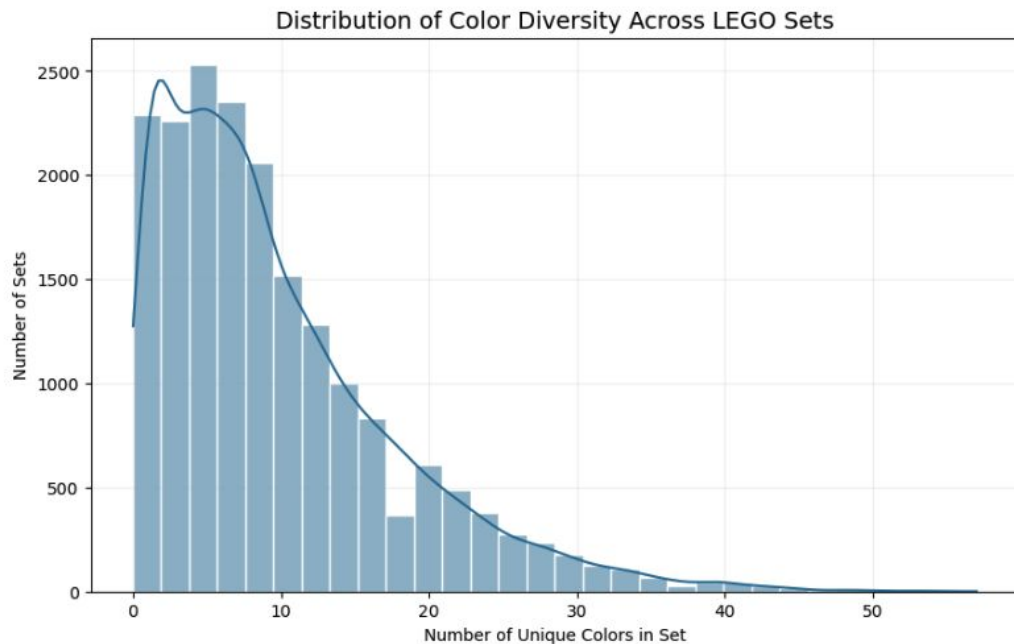
Large minifigures are rare:

75,159 Star Wars Death Star



DISTRIBUTION OF COLOR DIVERSITY

**MOST SETS FALL WITHIN
A MODERATE RANGE OF
COLOR DIVERSITY**

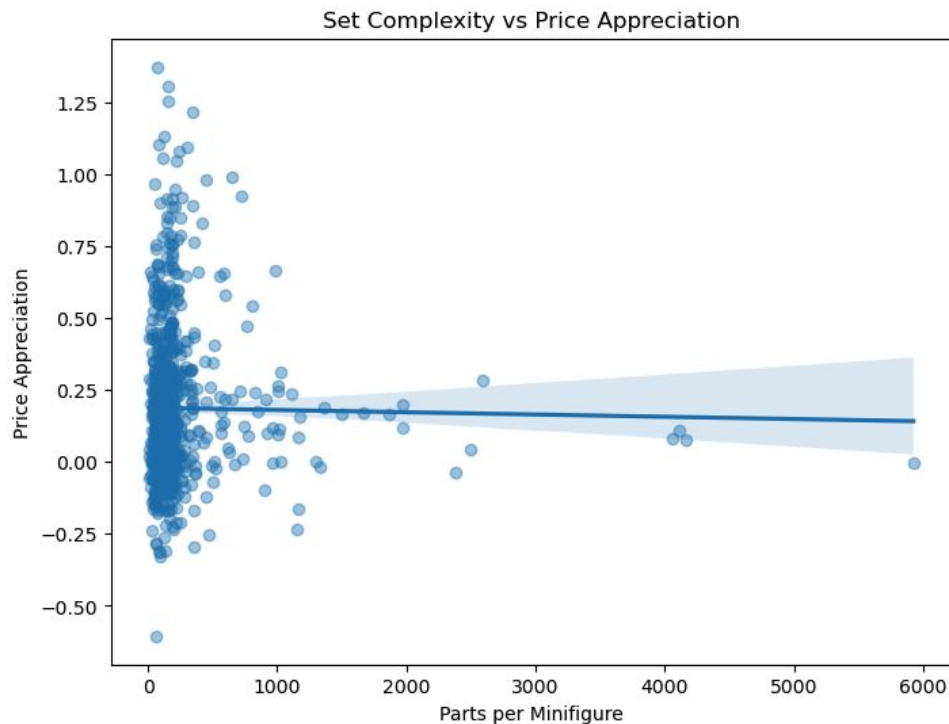


This suggests LEGO uses controlled color palettes, likely for:

- design coherence
- manufacturing efficiency
- theme consistency

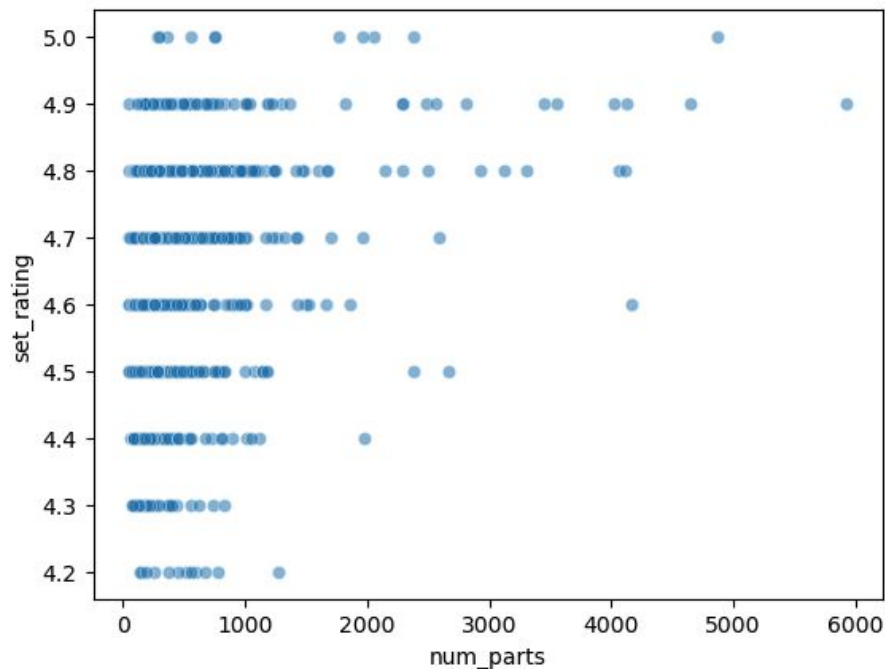
SET COMPLEXITY VS PRICE APPRECIATION

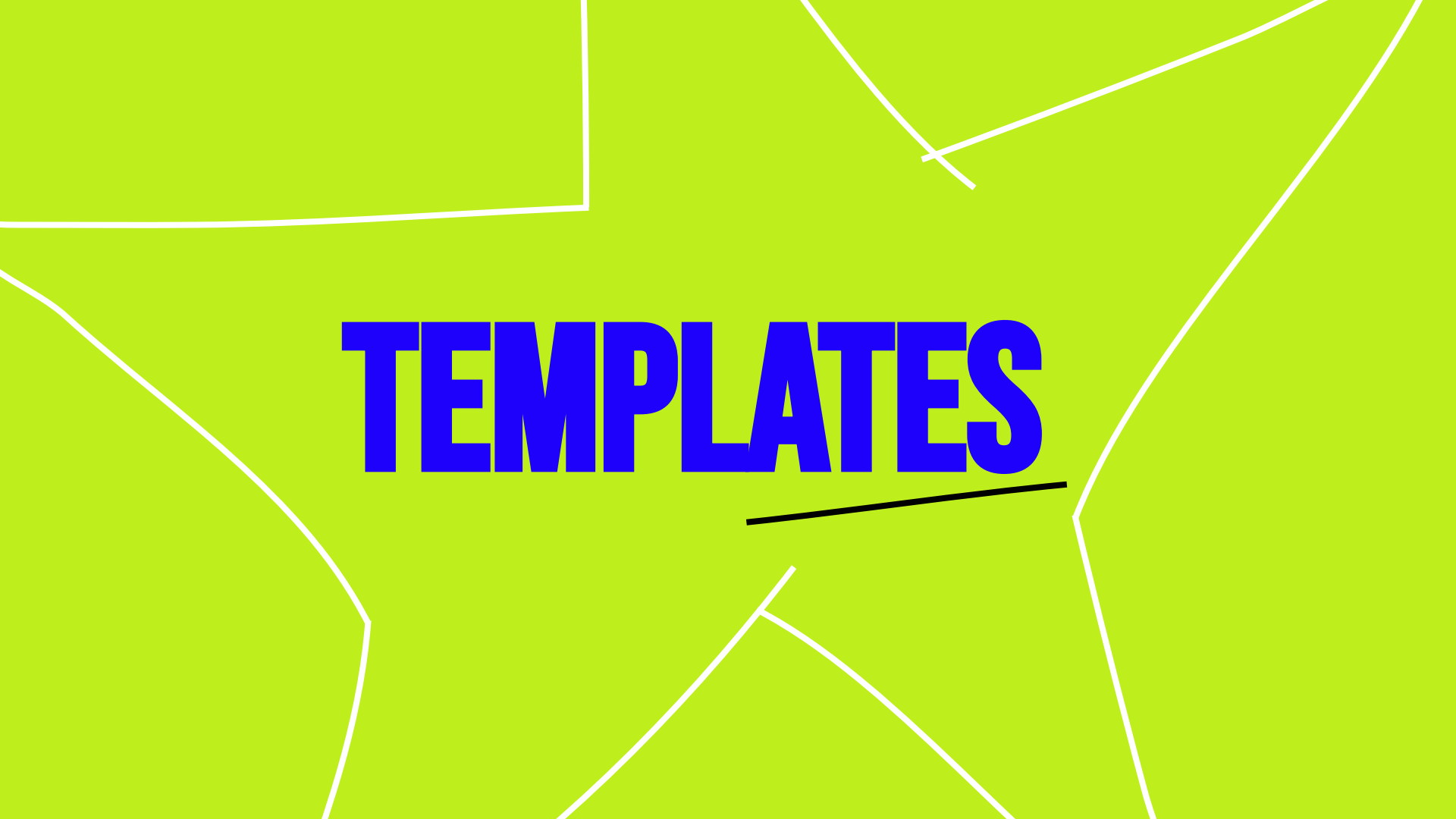
**SET COMPLEXITY DOES
NOT STRONGLY PREDICT
PRICE APPRECIATION.**



SET SIZE VS RATINGS

**BOTH SMALL AND
LARGE SETS CAN
ACHIEVE VERY HIGH
RATINGS**



The background is a solid yellow color with several white lines of varying thickness and orientation. Some lines are straight, while others are curved. They intersect and overlap, creating a dynamic, abstract pattern. The word 'TEMPLATES' is centered in the middle of the image.

TEMPLATES



HELLO, MEET YOUR HOST!

**HOST NAME
HERE**

Lorem ipsum dolor sit amet,
consectetur adipiscing elit



PLAYER 01

PLAYER NAME
HERE

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PLAYER 02

PLAYER NAME
HERE

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PLAYER 03

PLAYER NAME
HERE

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HOW TO PLAY

- 01.** Lorem ipsum dolor sit amet, consectetur adipiscing elit
- 02.** Lorem ipsum dolor sit amet, consectetur adipiscing elit
- 03.** Lorem ipsum dolor sit amet, consectetur adipiscing elit
- 04.** Lorem ipsum dolor sit amet, consectetur adipiscing elit
- 05.** Lorem ipsum dolor sit amet, consectetur adipiscing elit
- 06.** Lorem ipsum dolor sit amet, consectetur adipiscing elit

HOUSE RULES



01. Lorem ipsum dolor
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02. Lorem ipsum dolor
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03. Lorem ipsum dolor
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The background is a solid yellow color. It is decorated with several white lines of varying thickness and orientation. Some lines are straight, while others are curved. They intersect and overlap, creating a dynamic, abstract pattern. The lines are primarily located in the upper and lower portions of the frame, leaving the center more open for the text.

LET'S PLAY

**LOREM IPSUM DOLOR SIT AMET, CONSECTETUR ADIPISCING ELIT,
SED DO EIUSMOD TEMPOR INCIDIDUNT UT LABORE ET DOLORE
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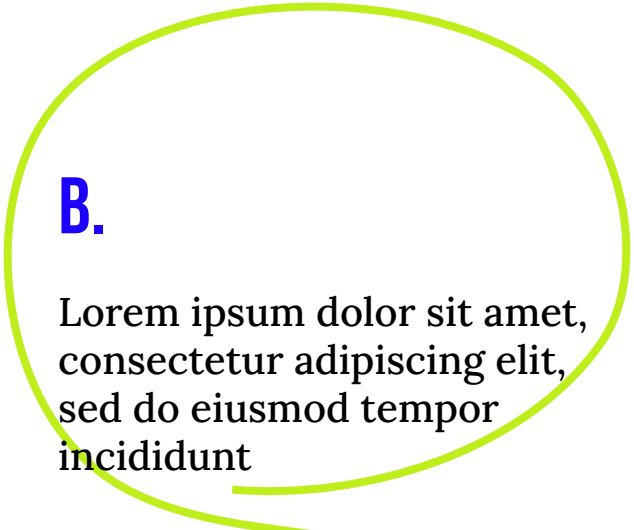
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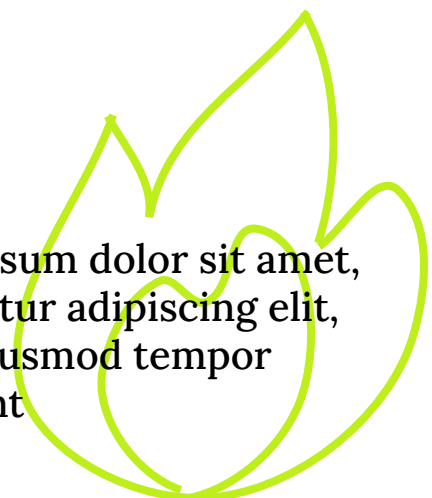
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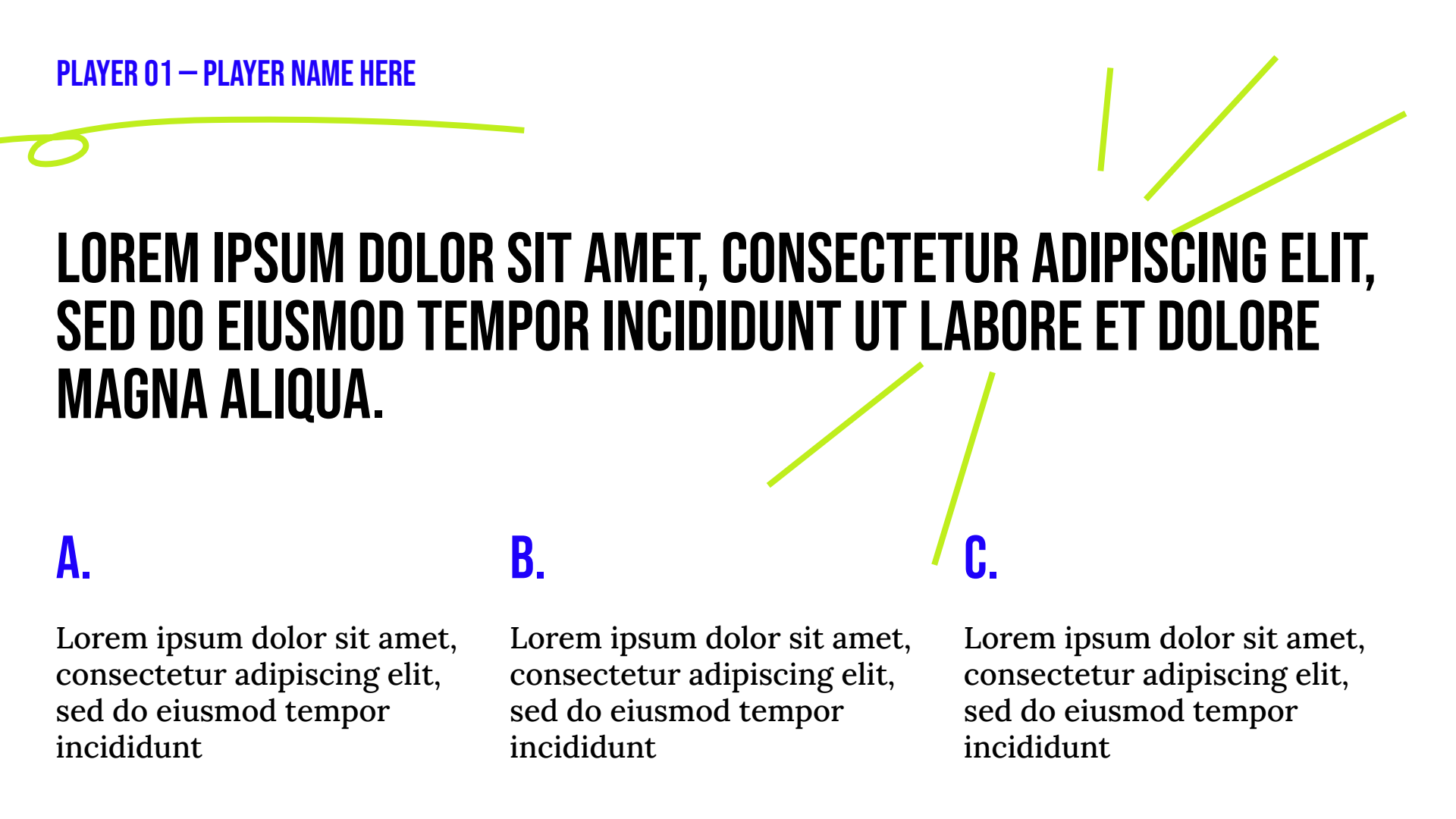
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PLAYER 01 — PLAYER NAME HERE



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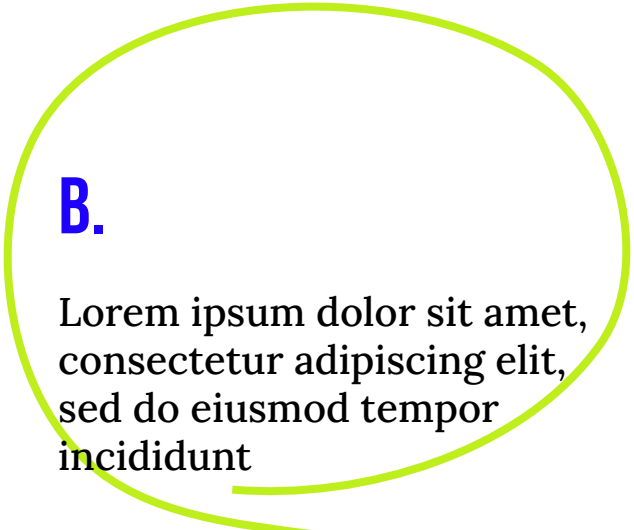
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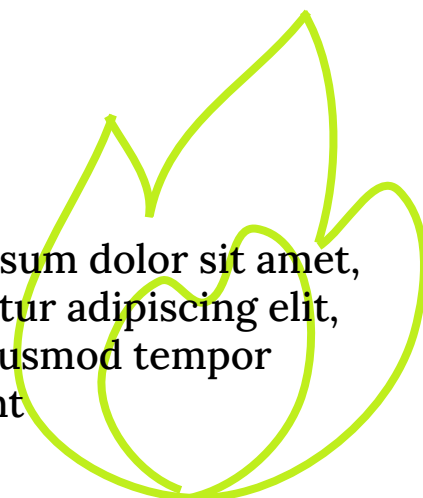
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**SUBMIT
YOUR VOTE**

FIRST PLACE



PLAYER 00

**PLAYER NAME
HERE**

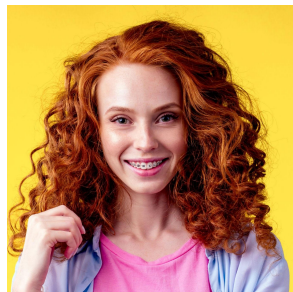
SECOND PLACE



PLAYER 00

**PLAYER NAME
HERE**

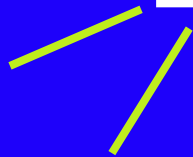
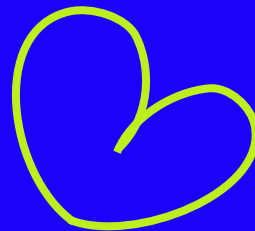
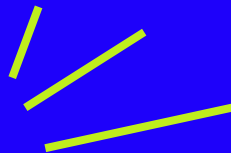
THIRD PLACE



PLAYER 00

**PLAYER NAME
HERE**

THANKS FOR
PLAYING



STICKER SHEET

